An Improved Integrated Model for Temporal Normalization of Satellite-Derived Land Surface Temperature

Wenhui Du, Xiangyang Liu, *Member, IEEE*, Zhao-Liang Li, *Fellow, IEEE*, Zhihao Qin, and Jinlong Fan

***Abstract*— Temporally incomparability across the scan lines in polar-orbiting satellite-derived land surface temperature (LST) affects their widespread application. Some challenges persist in the existing research on this issue, such as the absence of a universal algorithm applicable for the partly clear-sky condition in the daytime and scale inconsistency of the used datasets, when LST varies nonlinearly over time. Given this situation, we proposed an improved approach for temporal normalization, integrating ensemble regression models and a new LST variation rate model (RM), which captures typical LST variation characteristics over time during polar-orbiting satellite overpass periods. The Aqua Moderate Resolution Imaging Spectroradiometer (MODIS) LST data across the contiguous United States (CONUS) were collected to investigate its effectiveness. Moreover, cross-validation was conducted using the time-interpolated Geostationary Operational Environmental Satellite 16 (GOES-16) advanced baseline imager (ABI) LST. The normalized LST had remarkable consistency with the GOES-16 LST, with superior accuracy in contrast with the original LST. The root-mean-squared error (RMSE) was improved by approximately 1.56 K, and bias was enhanced up to 1.80 K. This study exhibited relatively superior performance in terms of quantitative outcomes and spatial distribution of LST compared with the previous studies. These evaluations indicate that the proposed method could be a dependable and general solution for addressing temporal inconsistencies in clear-sky LST during polar-orbiting satellite overpass periods.**

***Index Terms*— Aqua Moderate Resolution Imaging Spectroradiometer (MODIS), gradient-boosting regression tree (GBRT), land surface temperature (LST), temporal consistency, temporal normalization.**

I. Introduction

LAND surface temperature (LST) reflects the process of surface energy and water balance exchange from local to global scales [1], [2]. Its significance as a fundamental aspect of climate and the environment in geoscience has been recognized [3], [4]. Various algorithms [5], [6] have been developed to retrieve LST from thermal infrared (TIR) data observed by satellites. LST products derived from multiple satellites are generated using these algorithms, such as the MOD/MYD LST products from the Moderate Resolution Imaging Spectroradiometer (MODIS) onboard the Terra and Aqua platforms. These products have been systematically validated and exhibit accuracy within 1 K [7], [8], [9], [10], [11]. Despite their outstanding performance, these products suffer from the drawback of temporal incomparability for pixels along the scan lines, resulting from the design observation characteristics and large field of view of sensors in polar orbit satellites. Specifically, the observation time (local solar time) varies significantly for different pixels within the same image or the same pixel from images at different dates during a revisit period [1], [2], [12], [13]. This limitation affects the applications of multiple scenes of LST images at different locations, especially for a broad spatial extent, regardless of whether these scenes are from the same sensor or different sensors.

Much effort has been devoted to mitigate the temporal effect of LST products [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. Physically based diurnal temporal cycle (DTC) models [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24] are the main focus. For instance, the typical DTC pattern dealt with the National Oceanic and Atmospheric Administration—Advanced Very High-Resolution Radiometer (NOAA-AVHRR) LST with orbit drift effect [16], [20] to produce LST at a consistent local solar time and temporal normalization in generating all-weather LST [21]. The DTC model has strict application conditions for clear skies throughout the daytime. In addition, their implementation for polar-orbiting satellite LST has disadvantages in that insufficient satellite measurements cannot derive the typical patterns of the diurnal LST cycle. Duan et al. [12] developed an empirical method specifically designed for the morning transit period of 10:00–12:00. Although this approach has shown promising results for MODIS Terra LST data, it is inapplicable for extended periods when LST varies nonlinearly over time, whether in the morning or afternoon. Zhao et al. [13] proposed a machine-learning-based method by obtaining the relationships between LST and other related parameters. These algorithms mainly concentrate on normalizing LST products during morning overpass periods.

Compared with the morning LST, LST in the afternoon presents greater challenges for temporal normalization because of the more complex variation characteristics and more significant impacts throughout a longer daytime period. Du et al. [25] proposed a hybrid and general methodology for both morning and afternoon overpass periods by combining

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regression models and a new LST variation algorithm, called the LST difference model (DM). Although the algorithm [25] has been successfully used in the normalization of the MODIS LST product over the afternoon overpass period, certain restrictions and uncertainties still exist when it comes to practical application. The DM proposed here was established by characterizing the LST differences between the normalized LST and the original observed LST variation over temporal differences between the observation and reference time. The model may have shortcomings, including sensitivity to data scale and dimensions, susceptibility to noise and outliers, and limited predictive ability. Therefore, in some cases, it is necessary to consider using other models to compensate for these shortcomings. On the other hand, the training dataset comes from point-scale ground measurements, whereas other variables come from satellite images or reanalysis data. The utilization of point-scale training datasets may lead to some uncertainties, such as the inability to fully represent the entire pixel due to heterogeneity and scale effects. These uncertainties can arise from the differences in spatial resolution between the training and prediction datasets, as well as the potential mismatch in representing the temporal changes in LST throughout the day. Therefore, it is important to carefully consider the limitations and potential biases associated with point-scale training datasets when applying them to larger scale analysis or prediction tasks. For the two limitations, it is essential to devote more efforts to develop general algorithms for the temporal normalization of LST.

In the abovementioned context, this study introduces an improved algorithm for the temporal normalization of LST at longer polar-orbiting satellite transit periods during the daytime. First, a new general LST variation rate model (RM) was proposed by considering the relative advantages in terms of scale invariance, trend capture ability, and robustness by being relatively insensitive to outliers and noise. Furthermore, to account for scale effects, scale-consistent training and prediction datasets are generated mainly from stationery and MODIS products to predict corresponding parameters through ensemble regression algorithms. The methodology's effectiveness was examined using Aqua-MODIS LST products (MYD11) over the afternoon overpass period, involving intercomparison with a previous study [25].

The remainder of this article is structured as follows. Section II describes the study area and datasets used in this study. Section III presents the methodology scheme, including the development of a general LST variation RM over time and its combination into the ensemble regression models. Section IV gives the normalized results and corresponding discussion, and conclusions are drawn in Section V.

II. Study Area and Data

The proposed methodology was tested in the study area of the contiguous United States (CONUS). Due to the vast spatial extent, approximately 4500 km from east to west and 2700 km from north to south, significant temporal in-comparability existed in polar-orbiting satellite-derived LST products within the study area. More details are shown in [25].

This study employed multiple datasets, such as MODIS, Geostationary Operational Environmental Satellites (GOES)-16, ASTGDEM, in situ Surface Radiation Budget Network (SURFRAD) data, and other datasets, for the temporal normalization of LST. A detailed description of each data type and their respective preprocessing steps is presented in the following part of this section. Their characteristics are presented in Table I. The experimental polar-orbiting LST to test the proposed method was obtained from the MODIS LST dataset-MYD11A1 [6] onboard the Aqua satellite with equatorial transits at approximately 13:30 in ascending orbit, which was set as the standard time for normalization over the afternoon overpass period. Satellite-observed GOES-16 LST with a high temporal resolution of 1 h was employed to generate the RM parameters. In addition, the temporal interpolation method was adopted to obtain temporal consistency of GOES-16 LST for intercomparison with the normalized LST. Moreover, the view zenith angle (VZA) and observation time are extracted from these LST datasets as input terms of the RM model for calculating normalized LST and as conditions layers of pixel pairs filtering for validation.

Furthermore, the remaining datasets were used for extracting features in training and predicting datasets for computing normalization parameters, which involved day of year (DOY), total precipitable water (TPW), normalized difference vegetation index (NDVI), solar zenith angle (SZA), latitude, DEM, slope, aspect, and albedo.

A. Normalized Difference Vegetation Index

MYD13A2 products with a temporal resolution of 16 days were used to extract daily NDVI from linear interpolation for prediction purposes. While the GOES-16 NIR (channel 3-ABI L1b RadC03) and red (channel 2-ABI L1b RadC02) advanced baseline imager (ABI) L1b radiance data were utilized to obtain NDVI data in the training phrase according to the GOES-16 Product Definition and Users' Guide (PUG; pp. 27–28). Notably, these calculated NDVI values were instantaneous and varied within the daytime, whereas daily NDVI values were necessary for training. Therefore, the midday hour window average method [26], [27] was used to composite daily NDVI values from a 4-h time window centered at solar noon from 10:00 h A.M. to 2:00 h P.M.

B. Albedo

ABI L2 level LSAC products were processed in the training phase to derive daily ALBEDO from the average values of their high-temporal observations within a 4-h time window [26], [27]. During prediction, the MCD43A1 Collection 6 dataset [28] at a 500-m spatial resolution was used to extract daily ALBEDO.

C. Total Precipitable Water

ABI-L2-TPWC products were collected to obtain the daily TPWC variable at the reference time by temporally linear interpolation from adjacent observations. It is worth mentioning that due to time inconsistency issues in pixels along the scan lines of MODIS water vapor, the TPW data for predicting were also used this data.

In addition, land surface long-wave radiation datasets in 2020 with a 3-min temporal resolution from seven SURFRAD sites were collected to derive LST [25]. These LST data were

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TABLE I
MAJOR CHARACTERISTICS OF DATASETS USED IN THIS CONTEXT

Datasets	Training datasets (resampled to 0.02°, duration:2021231-2022230)		Prediction datasets (resampled to 0.01°, duration:2020001-2020366)	
	Source products	Spatial/temporal resolution	Source products	Spatial/temporal resolution
LST [6]	ABI L2 LSTC	0.02°/hourly	MYD11A1	0.01°/daily
NDVI [26-27]	ABI L1b RadC02 ABI L1b RadC03	0.005°/5minutes 0.02°/5minutes	MYD13A2	0.005°/16-day
DOY	/	0.02°/daily	/	0.01°/daily
Latitude	/	0.02°	/	0.01°
SZA	/	0.02°/daily	/	0.01°/daily
DEM, ASPECT, SLOPE	ASTGDEM_v3	0.0003°	ASTGDEM_v3	0.0003°
TPW	ABI L2 TPWC	0.1°/5minutes	ABI L2 TPWC	0. 1°/5minutes
Albedo [26-28]	ABI L2 LSA	0.02°/5minutes	MCD43A1	0.01°/daily
Normalization parameters	ABI L2 LSTC Observe time	0.02°/daily	MODIS LST Observe time	0.01°/daily

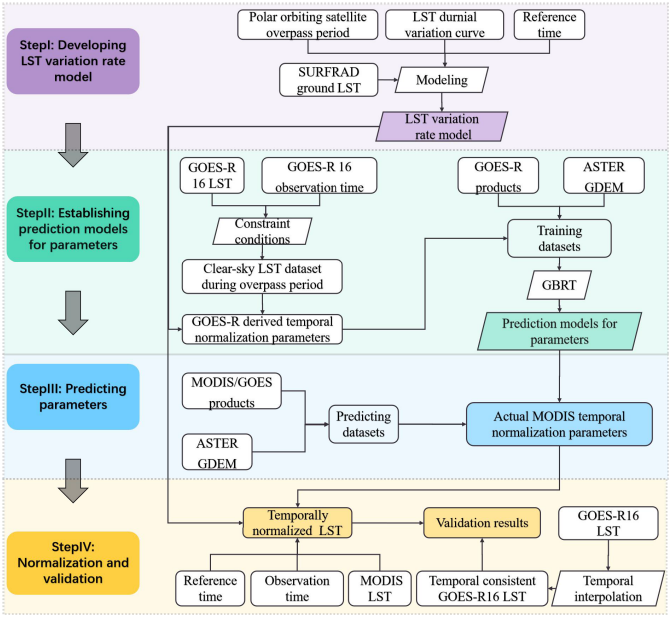

Fig. 1. Flowchart for temporal normalization of polar-orbiting satellite-derived LST based on geostationary observations.

used for modeling, along with GOES-16 LST data of high-temporal resolution. More details are shown in Table I.

III. METHODOLOGY

Fig. 1 illustrates the flowchart of the developed methodology, comprising four significant steps. First, in Section III-A, we establish the LST variation RM. Second, in Section III-B, ensemble regression models were trained to obtain prediction models for the parameters of the LST variation RM. In Section III-C, the actual temporal normalization parameters for polar-orbiting satellites were predicted. In Section III-D, the temporal normalized LST data were calculated from the LST variation RM and the corresponding actual temporal normalization parameters. In addition, intercomparison with the existing methodology [25] was performed with temporal-consistent GOES-R16 LST.

A. Developing a General LST Variation Rate Model for Polar-Orbiting Satellite-Derived LST

By considering the relative advantages in terms of scale invariance, trend capture ability, and robustness by being relatively insensitive to outliers and noise, a general LST variation RM for both morning and afternoon polar-orbiting overpass periods was proposed.

Compared with DM, which focuses on the absolute temporal change amplitude of LST, RM mainly focuses on the relative variation rate of LST between different times. Specifically, to capture the temporal changes in LST during daily polar-orbiting overpass periods and normalize polar-orbiting LST, two definitions associated with LST and time were initially introduced, named to LST variation rate (T_{ratio}) and temporal difference (t_δ). Among them, t_δ represents the temporal differences between the actual observation time and the reference time set for normalization. T_{ratio} quantifies the magnitude of LST changes occurring during these specific time intervals. The definitions of these terms are as follows:

$$T_{\text{ratio}} = (T_{\text{ac}} - T_{\text{norm}})/T_{\text{norm}} \tag{1}$$

$$t_\delta = t_{\text{ac}} - t_{\text{norm}} \tag{2}$$

where T_{ac} and T_{norm} infer the LST before and after normalization, respectively. t_{ac} is the actual observation time before normalization, and t_{norm} indicates the reference time for normalization, a constant value of the local solar time, typically set at 13:30 h (10:30 h) during afternoon (morning) transit periods. Time is local solar time.

Furthermore, this study investigated the variation characteristics of T_{ratio} over t_δ under clear-sky conditions within the overpass periods based on the high-temporal observed LST data from SURFRAD sites and the GOES-16 satellite. In the process, the polar-orbiting satellite overpass periods were set between 12:00 and 15:00 h (9:00–12:00) in the afternoon (morning). Correspondingly, t_δ ranges from −1.5 to 1.5 h. Fig. 2 illustrates the patterns of T_{ratio} variation over t_δ, which demonstrate evident quadratic curve shapes during the specific polar-orbiting satellite overpass periods. Consequently, T_{ratio}

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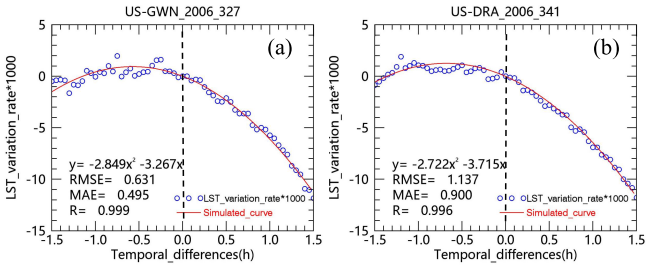

Fig. 2. LST variation rate distribution over various temporal differences (the observation time minus the reference time) from SURFRAD sites. (a) Goodwin Creek (GWM, Woody Savannas), DOY 327, 2006. (b) Desert Rock (DRA, Barren), DOY 341, 2006. The weather condition is clear sky. Time is measured in hours and temporal differences that equal to 0.0 indicates the reference time (black dashed line).

can be expressed as follows:

$$T_{\text{ratio}} = at_\delta^2 + bt_\delta \tag{3}$$

where a and b demonstrate the quadratic coefficients for (3). When the temporal difference (t_δ) is 0, the variation rate (T_{ratio}) also equals 0. As a result, (3) passes through the point (0, 0), and thus, there is no constant term in (3). Fig. 2 also gives the simulation results of (3). The high accuracy, with low root mean squared error (RMSE), mean absolute error (MAE) values, and high Pearson correlation coefficients (R), demonstrates its effectiveness for capturing the LST variation over time. Eventually, the temporal normalized LST (T_{norm}) can be expressed as follows:

$$T_{\text{norm}} = T_{\text{ac}}/\left(at_\delta^2 + bt_\delta + 1\right) \tag{4}$$

$$T_{\text{norm}} = T_{\text{ac}}/\left[a(t_{\text{ac}} - t_{\text{norm}})^2 + b(t_{\text{ac}} - t_{\text{norm}}) + 1\right]. \tag{5}$$

As shown in (1)–(5), once parameters a and b are obtained, T_{norm} can be finally calculated. This calculation requires the known values of the polar-orbiting satellite observed LST and corresponding observation time.

B. Establishing the Prediction Models of Parameters

The prediction models of LST temporal normalization parameters a and b are developed by establishing complex relationships between these parameters and influential environmental factors that could affect LST. The gradient-boosting regression tree (GBRT) [29] was selected as the decoupled model for the enormous advantage in prediction efficiency and robustness shown in many studies [25], [30], [31].

Training datasets are essential to establish the parameter prediction models. In our previous study [25], station-based training datasets were created using normalization parameters extracted from ground-measured LST as the dependent variable and corresponding dependent variables derived from MODIS and Modern-Era Retrospective analysis for Research and Applications (MERRA) data at a point scale. As described in Section I, because of the heterogeneity and scale effects of sites, point scale data may not fully represent the entire pixel, leading to uncertainty in the normalization results. Therefore, this study chose to create a training dataset based on geostationary satellites with a resolution of 0.02°, which is more compatible with prediction datasets and has the same scale (0.01°) as MODIS data.

The dependent variables were identified as temporal normalization parameters, which were computed from the GOES-16 stationary satellite-derived LST with a high temporal resolution of 5 min by using RM. The applied dataset's time range was from DOY 231, 2021, to DOY 230, 2022. The clear-sky GOES-16 satellite-derived LST was initially filtered by simulating the nonlinear relationships between the GOES-16 LST and the corresponding observation times during the polar-orbiting transit periods. The relationships between the LST and time can be simulated as quadratic curves when it is clear sky during the polar orbiting transit periods. Furthermore, the constraints of quadratic coefficients below 0 and Pearson correlation coefficients above 0.95 were used to filter the clear-sky datasets during the transit periods [25]. The LST at the reference time was obtained by linear interpolating four adjacent two to four observations. Finally, the normalization parameters were derived as dependent variables from these filtered clear-sky LSTs based on RM.

While the independent variables were determined as influential factors that could affect the normalization parameter's variations, these factors comprise the atmospheric conditions, land surface characteristics, and solar radiation. Because RM shares the same components (observation time, reference time, and polar-orbiting satellite observed LST) and comparable unknown parameters with the LST difference method (DM) [25], the identical features used in the study [25] were chosen as independent variables in this research. These utilized features are DOY, TPW, NDVI, SZA, latitude, DEM, SLOPE, ASPECT, and Albedo, which were extracted from the corresponding GOES-16 products and ASTGDEM_v3 data, with a spatial resolution of 0.02°. Section II presents more specific details. Eventually, it was trained using 1 048 735 samples from the initial simulation results of the GOES datasets, spanning 2021230 to 2022231, all using a wide range of feature factors. The training dataset covers all kinds of CONUS situations and a diverse range for all the utilized feature factors. Additionally, most of the simulation R value is greater than 0.98 with high precision. These findings support the training process to produce accurate standardized LST data.

In the training process, 80% of the dataset was selected as the training datasets and 20% as test datasets. The hyperparameters used can be referred to in our previous study [25]. After training, the prediction models for parameters a and b were established and output for predicting actual satellite normalization parameters.

C. Prediction of Actual Satellite Normalization Parameters

This section aims at estimating the actual temporal normalization parameters of MODIS LST data, as depicted in Step III of Fig. 1. By using the temporal normalization parameter predicting models described in Section III-B, the actual temporal normalization parameters of MODIS LST data could be calculated when the prediction datasets were created and input into the models. Considering the minimal scale difference, the predicted parameter datasets (0.02°) were resampled into 0.01° universally by the nearest-neighbor method to be consistent with MODIS LST data. As described in Section II, the nine corresponding variables in the prediction datasets were mainly extracted from the datasets marked as predicting in Table I.

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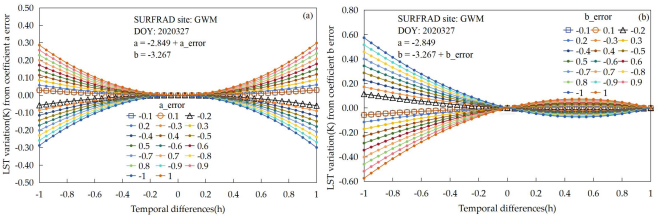

Fig. 3. LST variations (K) from (a) parameters a and (b) b errors for the proposed model.

D. Temporal Normalization and Evaluation Strategies

As in Step IV, the temporally normalized LST of polar-orbiting satellites was eventually estimated, combining the actual MODIS normalization parameters, RM, the actual observed LST, and observation time. Furthermore, these results were comprehensively validated. The proposed RM was intercompared with DM [25], concurrently implemented from the satellite- and station-based training datasets [25]. Therefore, the four methods (DM station-based, RM station-based, DM satellite-based, and RM satellite-based) were developed and cross-compared. The temporal-consistent GOES-16 geostationary satellite LST data were computed from the linear interpolation method for evaluating normalization results. For cross-comparison, normalized MODIS LST was resampled to 0.02° to match the time-interpolated GOES-R LST, as to mitigate the influences of the scale effects. Moreover, to illuminate angle effects in comparison, VZA constraints with differences less than 15° (for TDs within 1 h) or 20° (for TDs above 1 h) and a maximum of less than 50° were employed to filter pixel pairs of polar-orbiting satellite and GOES-16 LST for comparisons [25]. In addition, the pixel pairs of GOES-16 and MODIS with LST differences above 10 K were removed to achieve a more effective comparison and eliminate irrelevant impact factors. To directly examine the effectiveness of the proposed approach, visual inspection was first performed. Subsequently, quantitative analysis was conducted on four days in different seasons, with temporal differences exceeding 0.5 h. These specific time intervals were chosen because they could provide enough pixels and exhibited relatively substantial comparison improvements. Moreover, to explore the models' performance over various land cover types, RM with DM [25] located in different IGBP land cover types, three days randomly selected in each month. Finally, the importance percentages from permutation feature importance (MDI) were analyzed to identify the influences of features on temporal normalization parameters.

IV. Results and Discussion

A. Sensitivity Analysis for the Parameters in the Model

As mentioned, this approach takes specific feature factors as input datasets during the training phase, which may introduce certain limitations and challenges due to the uncertainty of these factors. They affect the normalization results through the GBRT model. Due to the complexity of this ensemble regression model, it is challenging to directly perform sensitivity analysis on the uncertainty of those feature factors. However, considering that they initially affect the accuracy of the parameters and then affect the normalization results, the direct sensitivity analysis of the parameters in the proposed model was conducted based on the existing simulation results. Fig. 3 illustrates a sample of the sensitivity analysis from the LST data of GWN site provided in Fig. 2(a). When the error of parameter a is -1.0 or 1.0 and b has no error, the LST variation is within 0.3 K. While when the error of parameter b is -1.0 and a has no error, the LST variation is within 0.6 K. These results demonstrate the robustness of our method.

B. Visual Variation Analysis

To demonstrate the performance of normalization methods in mitigating the temporal effects, Fig. 4 shows the comparisons for Aqua-MODIS LST daytime before and after normalization acquired on DOY 135, 2020. The missing pixels indicate that the land surfaces are overcast by clouds. As shown in Fig. 4(b), the observation time ranges from 11.8 to 14.1 h, and the responding temporal differences (the observation time minus the reference time 13:30 h) range from -1.70 to 0.60 h. In addition, an evident boundary in the observation time was located inside the masked subregion in a blue rectangle line. The temporal differences of the boundary's adjacent pixels in Fig. 4(h) were approximately 1.6 h, with the earliest observation time of approximately 12.4 h (in brown) and the latest of 14 h (in blue). Because of the vast temporal discrepancy, an obvious LST shift boundary can also be observed in the same masked subregion of the original LST images shown in Fig. 4(a) and (c). The discrepancy approximately reaches up to 4–6 K. Thus, this phenomenon confirms that it is essential for temporal normalization to obtain LST at a uniform local solar time.

Fig. 4(d)–(g) shows the corresponding normalized LST. Overall, this study successfully mitigated the noticeable LST discrepancy around the boundary, while preserving the initial thermal patterns associated with distinct surface conditions. According to the visual evaluation, the clear LST boundary almost disappears after normalization by the four algorithms because of the apparent temporal differences in the subregion. Regardless of using DM or RM, a comparable level of performance was found between the results from the same training datasets. Specifically, the satellite-based results in Fig. 4(f) and (g) exhibited better spatial consistency around the LST boundary than the station-based datasets in Fig. 4(d) and (e).

Fig. 4(i)–(l) gives the LST differences (the normalized LST minus the original LST), which present a spatial variation approximately from -3 to 2 K. Typically, slight LST variation is exhibited in the regions with observation time close to the reference time of 13:30, such as in the western and eastern sides of the study area. In contrast, the central region with relatively evident temporal gaps shows substantial LST differences. Similar to the spatial characteristic above, the LST differences from the same datasets were almost identical values over the study area, whether the DM or RM was applied. In addition, compared with the stationed-based datasets, LST difference maps from satellite-based datasets exhibited relatively significant absolute values around the boundary.

C. Quantitative Evaluation of Accuracy in Each Season

The scatter plots in Fig. 5(a) and (b), (e) and (f), (i) and (j), and (m) and (n) illustrate the comparisons of the

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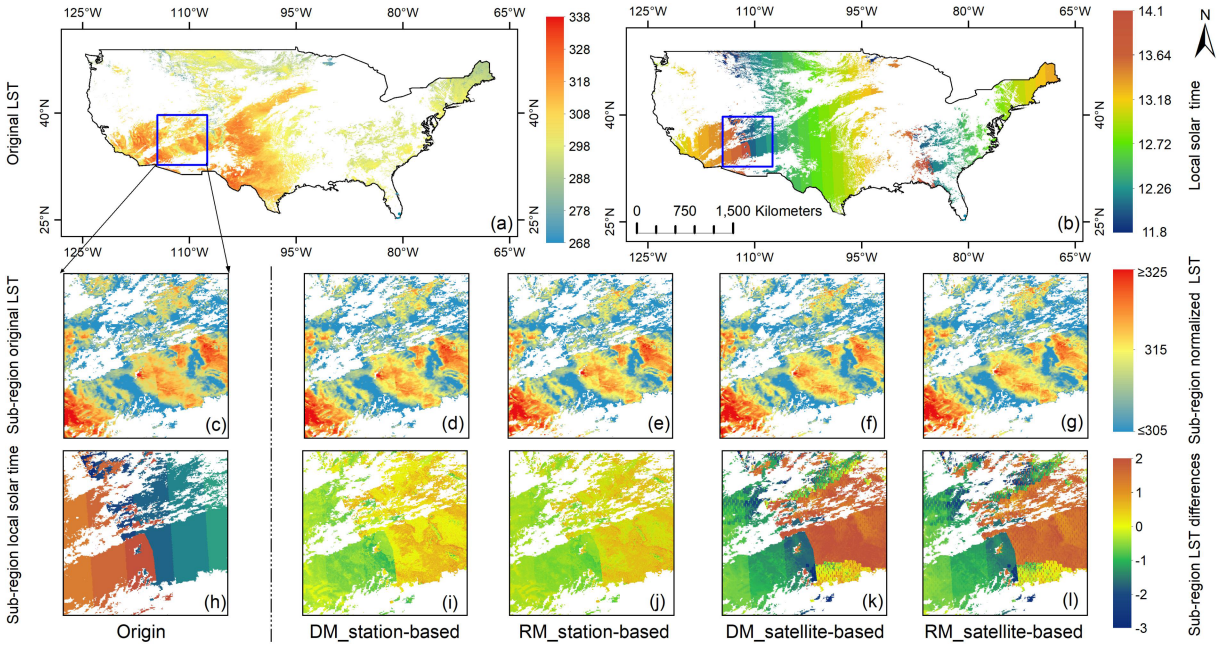

Fig. 4. Comparative visualization of temporal normalization results of Aqua-MODIS LST on DOY 135, 2020. (a) Original LST, blue rectangle is the subregion, (b) local solar time, measured in hours, (c) subregion original LST, (h) subregion local solar time, using the same color bar with (b), (d)–(g) subregion normalized LST by four methods, and (i)–(l) LST differences (the normalized LST minus the original LST, measured in Kelvins) by four methods.

original MODIS MYD LST and normalized LST at four dates in each season with temporal differences. Overall, the findings indicate that regardless of using DM or RM and whether training datasets used to predict their parameters were station-based or satellite-based, the accuracy from the temporally normalized LST is generally higher than that of the original LST. Fig. 5(c), (g), (k), and (o) displays the frequency distribution maps of variances between the temporally consistent GOES-16 and MODIS LST, before and after normalization, to further investigate the performance of the DM and RM methods implemented from station-based and satellite-based training datasets. Generally, the center of the histograms for the normalized LST shifted closer toward 0 compared to their distribution before normalization. Accordingly, the LST differences after normalization had lower mean values than those before normalization. This phenomenon corresponds to the tendency of scatter plots. As an illustration, after normalization, the mean difference (bias) values for DOY 009, 2020 in Fig. 5(c) and (d), change from −2.08 K (original results) to −0.99 K (DM station-based), −0.93 K (RM station-based), −0.52 K (DM satellite-based), and −0.69 K (RM satellite-based), respectively.

The mean difference value increased from 1.09 to 1.56 K. Similar tendencies can also be found in the results on other dates. The maximum bias value can reach up to 1.80 K (2020274).

Fig. 5(d), (h), (l), and (p) shows that the normalized RMSE values decrease from the initial value of 2.33 K to 1.16–1.39 K on DOY 009, 2020, from the initial value of 3.58 K to 2.20–2.75 K on DOY 114, 2020, from the initial value of 2.31 K to 0.93–1.55 K on DOY 274, 2020, and from 2.69 K to 1.93–2.02 K on DOY 297, 2020. Therefore, approximately 0.50–1.56 K of RMSE values were enhanced. In addition, temporal normalization for the four methods leads to significant reductions in the mean LST differences (bias) between the

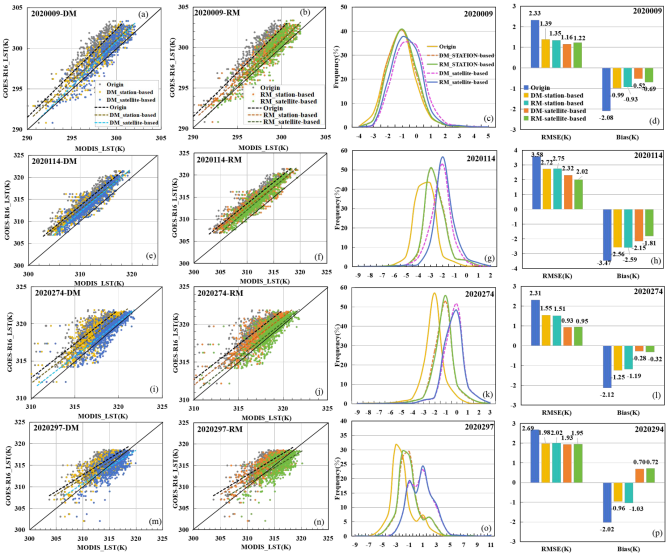

Fig. 5. Scatter plots between the original/normalized LST and temporal-interpolated GOES-16 LST at four specific dates. The dashed lines indicate the linear trend lines for each type. (a), (e), (i), and (m) represent DM, while (b), (f), (j), and (n) are for RM. The solid lines represent the center lines. (c), (g), (k), and (o) are frequency distribution of LST differences (the original or normalized LST minus the corresponding time-consistent GOES-16 LST). (d), (h), (l), and (p) present the statistical results between the original/normalized MODIS LST and temporal-interpolated GOES-16 LST at specific dates.

GOES-16 LST and MODIS LST. It was noticeable that when trained on the same datasets, the results generally had closer performances. Thus, regarding results from the four methods on the same data, discrepancies in precision were mainly dominated by training datasets. Furthermore, the RMSE and bias values from satellite-based training datasets have been substantially improved compared with those from the station-based training data. Thus, the accuracy of satellite-based results was

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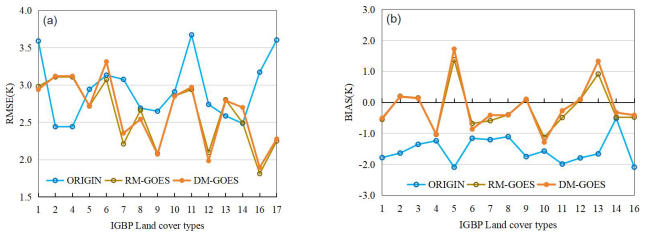

Fig. 6. Statistical indicators in various land cover types for (a) RMSE and (b) bias; units were Kelvin.

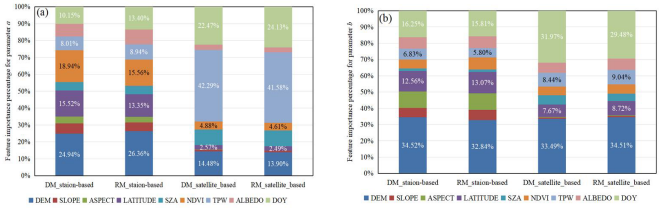

Fig. 7. Feature importance percentages for DM and RM results based on the station-based and satellite-based datasets. Result for (a) parameter a and (b) parameter b.

slightly superior to those of station-based ones, regardless of using the DM or RM technique.

Overall, it can be inferred that the study outcomes show a noteworthy enhancement compared with the results before normalization. When tested on the MYD11A1 product, the results from the satellite-based training data presented slightly better consistency and higher accuracy than the station-based training datasets. In addition, both the DM and RM techniques demonstrated similar levels of precision across different seasons. However, these tests on MODIS LST products did not reveal any significant advantages of the RM technique. When tested on other LST datasets that have more noise and greater variability, the DM and RM techniques may yield different results and exhibit unique strengths.

D. Analysis of Results for Various Land Cover Types

Fig. 6 presents the quantitative indicators of validation results through satellite-based RM and DM algorithms over various land cover types. Overall, the accuracy of normalized results from the two models was substantially enhanced after normalization with an identical degree over different land cover types. Regarding the RMSE values, the accuracy of normalized results from RM and DM was invariably higher compared with the nonnormalized data in a similar way. For bias values, relatively notable improvements in normalized LST from the initial LST were observed in most land cover types. At the same time, the most remarkable improvements reached 2 K. A comparable level of improved accuracy for RM and DM is presented among various land cover types.

E. Feature Importance Assessment

Feature (variable) importance refers to the features' contribution levels and the degree of usefulness to the model prediction. Thus, understanding the importance of features is crucial in interpreting and explaining the developed machine learning models. Fig. 7 gives the importance percentages of each explanatory feature of the prediction models. The results can be analyzed from three perspectives: models, parameters, and datasets.

Between different models: The results have demonstrated that features derived from the same datasets-based algorithms have identical important results, regardless of whether DM and RM were employed. This phenomenon is consistent with the evaluation results in Section IV-B, with both the DM and RM techniques demonstrating similar levels of precision. Considering the characteristics of DM and RM, they tend to provide comparable levels of accuracy over high-quality datasets. However, it is possible that when applied to other LST datasets with more noise and greater variability, it could yield distinct results and exhibit unique strengths.

Between different parameters: As quadratic coefficients in the DM and RM models, the difference between parameters a and b mainly lies in that they provide different ways of LST changes at different levels. Specifically, parameter a describes the curvature of LST changes over time, i.e., the acceleration of LST temporal changes. Features DEM and DOY contribute significantly to parameter a through the four algorithms, with weight values ranging from 32.84% to 34.52% and 15.81% to 31.97%, respectively.

In contrast, parameter b describes that the linear trend of LST changes over time, reflecting the rate of LST temporal change. DEM, NDVI, TPW, and DOY are the main contributing features for parameter b. The relationships between LST and DEM have been confirmed in previous studies [32]. Altitude increases would cause decreases in atmospheric pressure and density, allowing radiation to pass through the atmosphere easily to lower the LST. Specifically, the study area had a large elevation extent (approximately from 60 to 4000 m), which facilitates the impacts of elevation. Moreover, the annual variation in LST is also primarily influenced by DOY, a consequence of the Earth's revolution [33]. The Earth's elliptical orbit impacts solar radiation and atmospheric circulation, resulting in significant fluctuations in LST throughout the year. Furthermore, the two features of latitude and SZA have specific impacts on parameters that are closely related to solar radiation. These results were consistent with the previous study [12], [17]. The study [12] calculated the developed SLP from the empirical function established from DEM, NDVI, and SZA. In addition, the related analysis results showed that SLP was closely related to latitude.

Between different datasets: feature importance presents distinct results between different datasets. For parameter b, when it was derived from station-based algorithms, the most prominent weight values were from DEM, with 24.94% and 26.36% for DM and RM models, respectively. When it was derived from satellite-based algorithms, TPW owns the most significant weight values, with 41.58% and 42.29% for DM and RM models, respectively. Furthermore, for both parameters a and b, TPW makes greater contributions to the results from satellite-based algorithms than station-based ones. These results indicate strong correlations between parameters and TPW, which typically affects LST through greenhouse gas properties, cloud cover, and humidity [34]. This phenomenon may be because the satellite-based algorithms used the same scale-consistent training and predicting datasets, unlike the station-based algorithms.

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Overall, based on the abovementioned analysis and previous precision evaluations, the proposed algorithms have more advantages.

V. Conclusion

LST products derived from polar-orbiting satellites are frequently used as data sources in many fields. However, pixels along the scan lines are observed at different solar local times. The temporal incomparability hinders the usage of data collected over large areas and multiple satellites. This study demonstrated an integrated temporal normalization scheme for obtaining LST data at a consistent local solar time. The general LST variation RM was proposed for modeling clear-sky LST temporal variation characteristics during polar-orbiting satellite transit periods. Then, based on the high temporal advantage of geostationary observations, the ensemble regression model, the GBRT, was used to obtain the prediction models of the required parameters by establishing the relationships between the parameters and effective factors. Cross-comparison with the existing methodology [25] was conducted. The major conclusions are summarized as follows.

1) Spatial visual comparisons between the LST before and after normalization indicate that the temporally normalized LST shows a noticeable improvement to mitigate the temporal effect. When testing MODIS LST data, the results obtained from satellite-based training data showed superior consistency and higher accuracy than those from station-based training datasets. Both the DM and RM techniques demonstrated similar levels of precision. However, the results obtained from high-quality data may not fully show the strengths and differences between the two models.
2) According to the comparison statistics in each season, a considerable reduction in RMSE (up to roughly 1.56 K) and bias (reaching 1.80 K) can be detected for the normalized LST compared to the GOES-16 LST data. Similar improvements and accuracies of the satellite-based RM and DM algorithms were found in most cases.
3) For feature importance analysis, DEM and DOY mainly influenced parameter a. The main contributing features to parameter b were DEM, NDVI, TPW, and DOY, with DEM having the highest weight values for station-based algorithms and TPW having the highest weight values for satellite-based algorithms. Concurrently, TPW made significant contributions to satellite-based algorithms than the station-based algorithms for both parameters. The scale consistency of the training and predicting datasets may be the main reason.

Based on the evaluation provided, it can be concluded that the proposed method can illuminate temporal incomparability with reliability and stability. Once the required models are trained, the proposed method can be used for polar-orbiting satellite-derived LST products. In addition, in contrast to the earlier investigations, this method can normalize the LST during the afternoon overpass periods and applies to the data in the morning. Additionally, while this study primarily focuses on afternoon LST normalization using Aqua/MODIS, our upcoming research will center on normalizing full-daytime LST. Furthermore, exploring normalization using fewer input datasets is also another field for future study.

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